

Thermo-physical Characterisation of Plasters Containing Phase Change Materials (PCMs)

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Research Article

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Abstract

Phase change materials (PCMs) added into building materials largely demonstrated their effect in improving the overall thermal capacity, thus contributing to the reduction of the energy demand for heating and/or cooling. Knowing the thermo-physical properties of the enhanced materials is essential to maximize the achievable effect. The most used technique to do so is the Differential Scanning Calorimetry (DSC), but in case of inhomogeneous materials this method turns out to be poorly reliable, mainly due to the small amount of material usually needed to carry out the test. To overcome the main limitations, an alternative set up was realized, which allowed the characterization of different plaster samples enhanced with granular PCM. More specifically, plasters enhanced with 10% by mass of two different PCMs with melting temperatures of 28°C and 27°C, respectively, were considered. At first, a steady-state test was conducted and constant thermal gradients were maintained through which the thermal conductivity of the materials used was estimated. Then, an unsteady-state test was conducted through which the specific heat and the latent heat were estimated, showing a good agreement with values provided by the PCM suppliers. The estimated properties were then validated against experimental data acquired during the monitoring activity under real outdoor conditions of different wall samples on which the PCM-enhanced plasters were applied. With the estimated properties, RMSE values were lower than 1°C for temperatures and lower than 2.50 W/m² for heat fluxes.

1. Introduction

The achievement of carbon neutrality requires the reduction of the impact of the building sector, the most energy consuming sector. Buildings, in fact, are responsible for almost 40% of the total CO₂ emissions of European countries [1], whose building stock is strongly heterogeneous and changes at a very slow rate [2]. On average, only one building out of ten was built between 2000 and 2008, while almost one out of two has more than 50 years [3]. This consistent part of the actual building stock, mainly built before the introduction of any energy regulation, has poor energy performance, aggravated by old and obsolete equipment and appliances. The energy saving potential of these buildings is therefore huge. The IEA in 2017 estimated that deep energy renovations of existing building envelopes represent a saving potential greater than all the final energy consumed by the G20 countries in 2015, or around 330 EJ in cumulative energy savings to 2060 [4]. Interventions aimed at improving the energy performance of existing buildings are therefore mandatory in order to achieve not only the reduction of greenhouse gas emissions, but also the ambitious goals of reducing energy consumption for heating and cooling and improving indoor environmental quality. The main applied technologies include interventions on the building envelope, on the heating and cooling systems, on the lighting system, and the integration of renewable energy sources [5]. In order to achieve energy efficiency goals, multiple solutions can be combined. However, despite the consistent number of available technologies, several limitations hinder their application, such as design, complexity, integration, regulatory, economic or behavioural limitations.

In this context, phase change materials (PCMs) have been identified as one of the most interesting solution in building energy refurbishment. Being characterized by high energy storage capacity per unit

volume at a nearly constant temperature [6], PCMs act as an almost isothermal reservoir of heat [7], making buildings more energy effective [8]. When integrated in the building envelope, PCM increases the thermal mass of the building envelope while undergoing the phase change [9], thus allowing the storage of significant amount of energy without the uncomfortable temperature swings and the large structural mass that is necessary with sensible heat storage [10]. The remarkable effect a building envelope has on the energy demand for heating and cooling [11] led to much research over the years focusing on the integration of PCM within various building components [12–20]. PCMs can be directly integrated in the building structure, such as in bricks or concrete, or in cladding layers, such as in wallboards or plasters. The addition of PCM into the structural parts is possible only in case of new constructions, while the addition into finishing layers is possible in both new constructions as well as building renovations. As regards the focus of this research, PCMs were mixed within plaster, more specifically lime-based ones, considered more compatible with existing historical buildings. In order to maximise the effect of the PCM, the knowledge of the materials' properties is essential. In addition to this, when dealing with building materials the actual properties does not always agree with those declared [21]. This may be due to the fact that building materials are poorly characterized at laboratory scale, if considering sample sizing several centimetres [22].

The most used technique for the determination of the PCM thermophysical properties is the Differential Scanning Calorimeter (DSC) [23], which consists in the measure of the amount of heat required to warm up and cool down a sample of the material to be tested, which is estimated in comparison to that of a reference material. However, this technique is affected by some critical limitations due to the sample's properties. In a DSC measure, in fact, the sample size is normally small, less than 15 mg, and has to be pure and homogeneous [24]. Nevertheless, in case of building envelope applications, PCMs are usually added as encapsulated materials, both micro-encapsulated, in the form of powder or granules, as well as macro-encapsulated, such as within plastic or metallic containers. The characterization of a building material enhanced with PCM, therefore, cannot be considered reliable if carried out through a DSC analysis. To overcome these limitations, an alternative set up was considered necessary in order to roughly estimate the new enhanced materials' thermal properties. Several researchers had already tried in the past to overcome the main instruments' limitations by developing their own set up.

Yesilata and Turgut [25], for instance, developed a dynamic measurement technique to measure the effective thermal properties of anisotropic materials consisting in a dynamic adiabatic-box where the outer and bottom walls were heavily insulated while the top wall was the test specimen, a thin layer whose thermal conductivity was much higher than that of the other walls. Inside the box, close to the bottom wall, there was a heater immersed into water and controlled by a thermostat that heats up water to a desired temperature. Once turned the heater off, the cooling rate of water could be considered a measure of the thermal transmittance of the specimen. This technique was found to be functional and robust for a preliminary estimation of thermal performances of anisotropic specimens, even though the testing time is quite long.

The challenge gets even harder when PCMs are integrated in the materials to be tested. As already mentioned, PCMs at lab scale are usually tested by means of DSC, which requires small and homogeneous samples, conditions that are hard to achieve. Several researchers worked for the development of new equipment able to use macroscopic samples, more similar to actual constructive systems. Among them, de Gracia et al. [26] developed a new equipment to test the steady and transient thermal response of building materials containing PCM. The set up consisted in a highly insulated wooden structure whose interior was divided into two cavities, which were meant to be used to simulate the inner and outer conditions of a building envelope by means of copper coils connected to programmable water baths. These were separated by the sample to be tested, which was placed between the two cavities with insulated edges to ensure the mono dimensional heat transfer. Three different types of experiments were possible to evaluate the performance of the test sample. The thermal conductivity was estimated in steady-state conditions using the thermal gradient between surfaces, then the heat storage capacity and the dynamic thermal response were estimated with unsteady-state tests. The former was determined by measuring the heat through the sample that was subjected to a determined temperature interval, while the latter was estimated by analyzing the delay between the temperature peaks reached in the cavities.

Barreneche et al. [27] at the University of Barcelona developed two devices to characterize the effective thermal conductivity and to register the temperature-time response curves of materials including PCMs at macroscale. The first one was a conductimeter, which consisted in a sample positioned between a hot and a cold plate, connected to thermostatic baths and both insulated to avoid any influence from the ambient temperature, surrounded by an insulation frame made of refractory bricks that allowed a mono-dimensional heat transfer on the z-axis. The thermal conductivity of the sample was determined through the thermal gradient achieved between the two surfaces, with an estimated accuracy close to 2%. The other device, named T-t curves device, consisted in two samples, a reference and the test one, positioned on the sides of a heating device and framed by an insulation casting to avoid heat transfer on the edges. During the experiments, which occurred at the controlled temperature of 18°C, the samples were heated until a steady-state condition was reached and then, once the heating plate was disconnected, they were thermally compared during the cooling process by means of several temperature sensors.

Borreguero et al. [28] developed an experimental set up to evaluate the improvement of the thermal storage capacity of gypsum samples integrated with different percentages of PCMs. It consisted of a hollow aluminium box through which water, at a determined temperature, flowed continuously by means of a peristaltic pump, connected to a thermostatic bath. Each of the samples to be tested was placed on the surface of the aluminium box and then wrapped up in insulation. Two different studies were carried out, namely an absorption study, where the temperature of the thermostatic bath was changed from 21°C to 40°C, and a reversibility study, where the temperature of the thermostatic bath was cyclically changed from 21°C to 36°C.

Through the customised set up realised at the University of Ferrara [29], the experimental characterisation of two PCM-enhanced plasters was carried out, followed then by the thermo-physical properties

validation against experimental data gathered under real outdoor conditions. The PCMs used have phase change temperatures of 27°C and 28°C, and are hereafter named TK27 and AS28, respectively.

2. Methodology

2.1. Thermo-physical properties estimation

At first experimental tests inside the climatic chamber were carried out, aimed at characterizing the plaster's thermal properties, namely the thermal conductivity, the specific heat and the latent heat capacity, using the set up shown in Fig. 1(a). Three plaster samples were realised, a lime-based one used as reference, and two in which 10% by mass of PCM were added, one with a phase change temperature of 27°C and one of 28°C, purchased from TITK [30] and Smart Advanced Systems GmbH [31], respectively, shown in Fig. 1(b). The thermo-physical properties were estimated following the methodology previously reported in [29]: the thermal conductivity was estimated through a steady-state test while the specific and latent heat were estimated through a two-step ramp unsteady-state test, as shown in Fig. 2.

2.2. Properties validation

After the tests carried out at lab scale, a bigger experimental set up was designed and realized to simultaneously monitor different wall samples. On one side, this allowed the monitoring of the behaviour of the PCM-enhanced plasters under real weather conditions, on the other the validation of the estimated properties against experimental data. The installation of the set up was done exploiting an already existing mock up facility (Fig. 3(a)) at the TekneHub Laboratory at the University of Ferrara, which was built in 2016 within the European project LIFE Climate Change Adaptation – HEROTILE (High Energy savings in building cooling by ROof TILE shape optimization toward a better above sheathing ventilation) [32].

The set up realized has an aluminium structure covered by OSB panels, and was designed to test up to four wall samples at the same time. The set-up is approximately 0.90 m x 0.90 m x 0.50 m, while each available slot, containing the wall samples, is 0.38 m x 0.38 m x 0.50 m Fig. 3(b). Each wall sample realized consists of a brick layer covered with plaster both on exterior as well as on the interior side. However, in order to ensure the maximum flexibility and the possibility of changing the configurations to be tested, the wall samples are made of two identical parts, consisting in a 0.03 m masonry tile layer covered by 0.03 m of plaster, which were kept together by means of wooden frames and metal profiles, as depicted in Fig. 4. The wall samples were 0.30 m x 0.30 m and had a 0.04 m insulation frame to limit the heat transfer on the edges, thus establishing a mono-dimensional heat

transfer through the walls and prevent them from influencing each other. T-type thermocouples (accuracy: 0.5 K) were positioned on the surface of the exterior and the interior plasters, as well as inside the mock up room to monitor air temperature. Heat flux meters with integrated temperature sensor (heat flux accuracy: 5%, temperature accuracy: 2% of value in °C) [33] were positioned between the two brick layers,

in the middle of each wall sample. As regards the boundary conditions, data in terms of air temperature, solar radiation, wind speed and wind direction were acquired by a weather station [34] already installed close to the mock up facility. Once the set up was put into operation, a numerical model was implemented in COMSOL Multiphysics v5.6 [35] through which the previously estimated thermo-physical properties were validated against the experimental data gathered. The four wall samples modelled and the physics implemented are depicted in Fig. 5 (where * are the COMSOL Multiphysics used). The validation of the model was carried out using the experimental data acquired over a few consecutive days, in which the outdoor conditions were in line with the season averages, both in terms of air temperature as well as solar irradiance. Given the simple model geometry the mesh implemented was made of regular hexahedrons. Meshes with different number of elements were tried and in all cases the element quality was equal to 1, hence no mesh independence of the solution was considered necessary and investigated.

3. Results and discussion

3.1. Thermo-physical properties estimation

At first, the thermal conductivity of the PCM-enhanced plasters was estimated. Data collected during the steady-state tests are reported in Fig. 6(a) for AS28 and Fig. 6(b) for TK27. The thermal conductivity was estimating using Fourier's law and by only using data when the system was in thermal equilibrium, hence when values of the two heat flux meters were almost identical. The thermal conductivity resulted from the test were $0.28 \text{ W/(m}\cdot\text{K)}$ for AS28 and $0.24 \text{ W/(m}\cdot\text{K)}$ for TK27.

As regards the unsteady-state tests, data collected during the two-step ramp are reported in Fig. 7(a) and Fig. 7(b). The detailed heating and cooling of AS28, comparing the phase change ramp (PC) to the no-phase change one (noPC), are depicted in Fig. 8(a) and Fig. 8(b), respectively. During heating as well as during cooling temperatures changed more slowly in correspondence of the phase change, with peak differences at the melting temperature both above and below the plaster sample. As regards heat fluxes, the peaks reached between the plaster sample and the masonry tile are much lower during heating with reductions of nearly 20 W/m^2 . Moreover, in correspondence of the phase change an inversion of the heat flux (HFmid) is clearly visible, during heating as well as during cooling. Then, detailed heating and cooling in case of TK27, comparing the phase change ramp to the no-phase change one, are depicted in Fig. 9(a) and Fig. 9(b). Analogous considerations to AS28 could be done: temperatures changed at a slower rate reaching peak

differences of 2–3 K while heat fluxes reached lower peaks, with reductions of about 30 W/m^2 during melting. In general, even though the PCMs quantities were quite limited, differences between the phase change and no-phase change ramps are clearly visible. As regards the melting of the PCM (20°C - 40°C ramp), in both the plasters a slowdown around 28°C was visible, while in case of the solidification of the PCM (40°C - 20°C ramp), a slowdown in the temperature change was still visible, but it was less clear before the phase change and also seemed to occur at a different temperature than melting, occurring at

around 25°C. To facilitate the comparison between the plasters containing PCMs, data collected during the tests were put together and HFmid heat fluxes were reported in Fig. 10, which were then compared to a reference plaster (REF). During melting, depicted in Fig. 10(a), the enhanced plasters reached lower peaks than the reference one, with reduction of about 62% for AS28 and 73% for TK27. As observed in the temperatures, where the sharp slowdowns occurred at almost the same temperature, the peak heat flux during the phase change (thus, the positive peaks)

were simultaneous, although reaching different peaks, equal to nearly 25 W/m² in AS28 and 30 W/m² in TK27. Another visible difference between the enhanced plaster regards the melting range of the

Table 1
Estimated thermo-physical properties of the PCM-enhanced plasters

	ρ [kg/m ³]	λ [W/(m·K)]	c_p [J/(kg·K)]	Δh [kJ/kg]
REFERENCE	1646	0.31	895	-
AS28	1522	0.28	1072	6.8
TK27	1365	0.24	1064	10.8

PCM, which seemed to be narrower in case of AS28 than TK27. Different considerations emerged comparing the heat fluxes during solidification, depicted in Fig. 10(b). Both AS28 and TK27 reached lower peaks than the reference one, with reductions of about 30% and 28%, respectively, but the greatest difference regards the phase change peak temperature, which seemed to happen at lower temperatures than melting and also different ones. The first to occur, hence at a higher temperature, was in AS28 and then in TK27. From the data obtained during the two-step ramp the latent heat capacity of the plasters containing PCM were estimated, resulting in 6.8 kJ/kg for the plaster containing AS28 and 10.8 kJ/kg for that containing TK27. Considering the amount of granular PCM contained in each of the enhanced plasters, the latent heat capacity of the granules only was estimated to be about 92 kJ/kg for AS28 and 145 kJ/kg for TK27. As regards the specific heat, only data gathered from the no-phase change ramp were used. Calculating the cumulative heat stored and released during heating and cooling, respectively, as shown in Fig. 11, and considering each plaster sample mass and the total temperature difference, the values estimated were 1072 J/(kg·K) for AS28 and 1064 J/(kg·K) for TK27. The estimated thermo-physical properties of the two PCM-enhanced plasters according to the data previously reported are summarized in Table 1.

3.2. Properties validation

The model of the experimental set up, described in paragraph 2.2, was validated against the experimental data collected. More specifically, a period of four consecutive days was selected, the first of which was

used only to bring the system up to regime, thus excluding the effect of the initial

conditions on the results. For this reason, the results of the model validation, that was carried out with a timestep of 10 minutes and are depicted in Fig. 12(a) for AS28 and Fig. 12(b) for TK27, cover a period of three days. The results showed a good agreement between the experimental and the simulated data, supported also by the root mean square error (RMSE) calculated as in Eq. (1) and further reported in Table 2:

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N}}$$

1

where y_i is the actual measured value and $\hat{y}_i$ is the simulated one.

In terms of temperatures, RMSE values are always lower than 1°C, with the highest values equal to 0.89°C and 0.87°C in case of T_{out} , namely the temperature on the outer surface of the wall samples. As regards the other temperatures, RMSE values are even lower and between 0.48°C and

Table 2
Root mean square error (RMSE) values between
experimental and simulated data

RMSE	T_{out} [°C]	T_{mid} [°C]	T_{in} [°C]	HF [W/m ²]
AS28	0.87	0.48	0.57	1.90
TK27	0.89	0.59	0.56	2.40

0.59°C. In term of heat fluxes, RMSE values are 1.90 W/m² for AS28 and 2.40 W/m² for TK27. All these values indicate a good agreement between the experimental data, collected during the real scale monitoring, and those simulated with the model implemented in COMSOL Multiphysics whose thermophysical properties were the ones experimentally estimated by means of the ad hoc set up.

4. Conclusion

Phase change materials (PCMs) integrated within building construction materials represent a possible strategy for the energy refurbishment of existing buildings, with special regards to buildings located in particular contexts in which many of the most common techniques are forbidden. To maximise the effect of PCMs on building performance, the knowledge of the thermo-physical properties of the new enhanced materials is of outmost importance. However, building materials with PCMs added within are strongly inhomogeneous and standard tests for their thermo-physical characterisation are not suitable. To overcome these issues, an alternative set up was designed and realised ad hoc at the TekneHub laboratory of the University of Ferrara in order to estimate, with different tests but one configuration only,

the thermal conductivity, specific heat and latent heat capacity of plasters containing PCMs under the form of granules. The values estimated through steady-state as well as unsteady-state tests were then validated against experimental data. Another experimental set up was in fact designed and realised in order to monitor the behaviour of up to four wall samples under real outdoor conditions: this allowed not only the monitor of the enhanced plaster over long periods under different conditions, but also the validation of the previously estimated properties. A numerical model of this set up was implemented in COMSOL Multiphysics and simulated data were compared to the experimental ones, and results showed good agreement between the two, with root mean square error values lower than 1°C and 2.50 W/m² for temperatures and heat fluxes, respectively. These results support the reliability of the alternative set up designed as a tool for the thermo-physical characterisation of composite and inhomogeneous materials, being able to overcome the limitations of standard devices and estimate multiple properties with one configuration only.

Nomenclature

Δh	latent heat capacity [kJ/kg]
AS28	granular PCM with phase change temperature at 28°C
TK27	granular PCM with phase change temperature at 27°C
PC	phase change
noPC	no phase change
RMSE	root mean square error

Declarations

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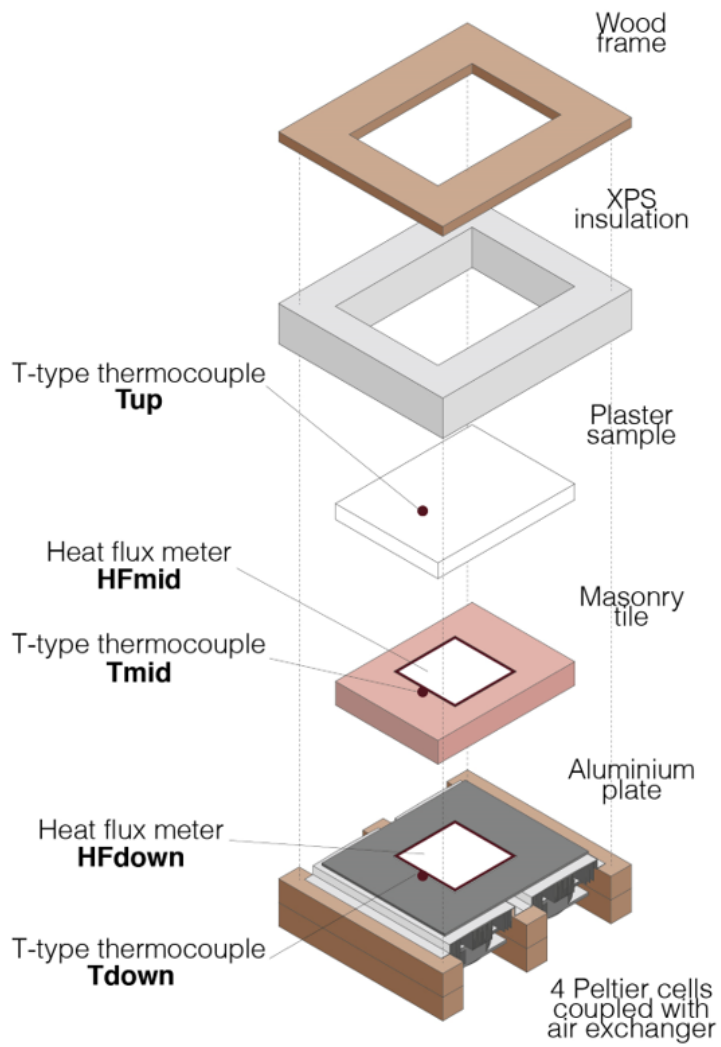
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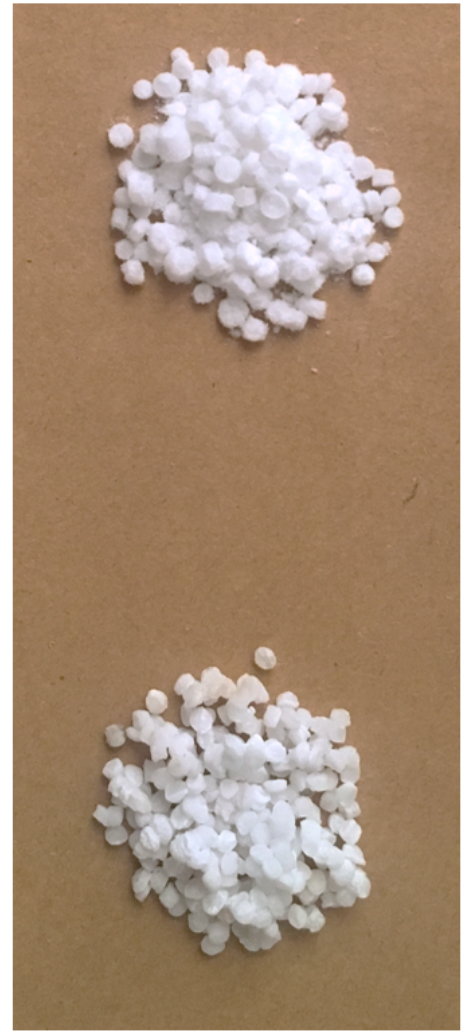
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Figures



a



b

Figure 1

(a) Materials and sensors' position of the experimental set up; (b) PCMs used: top AS28, bottom TK27

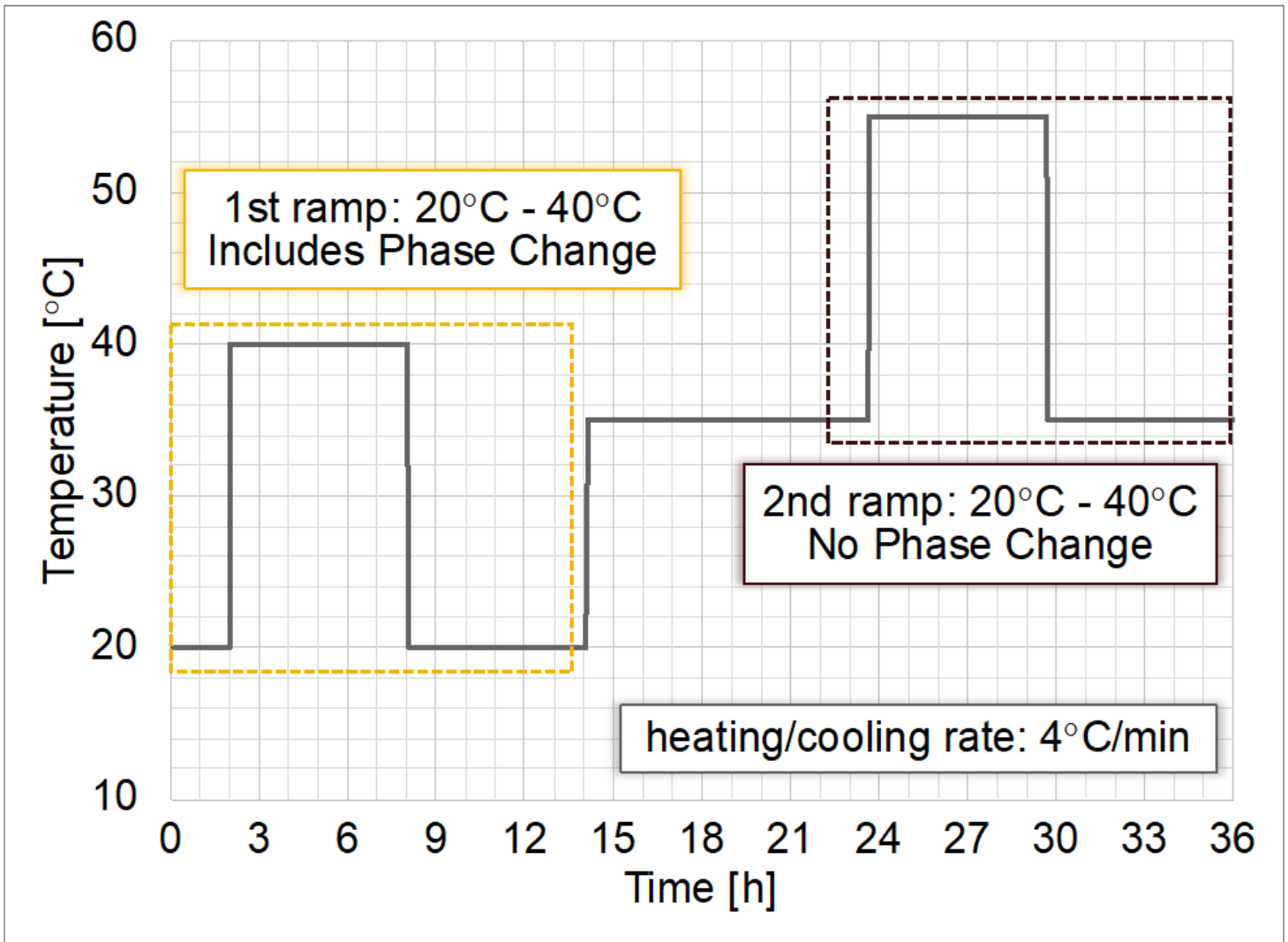


Figure 2

Temperature set inside the climatic chamber during the unsteady-state test



a



b

Figure 3

Mock up at the TekneHub Laboratory of the University of Ferrara, (a) before and (b) after the installation of the set up

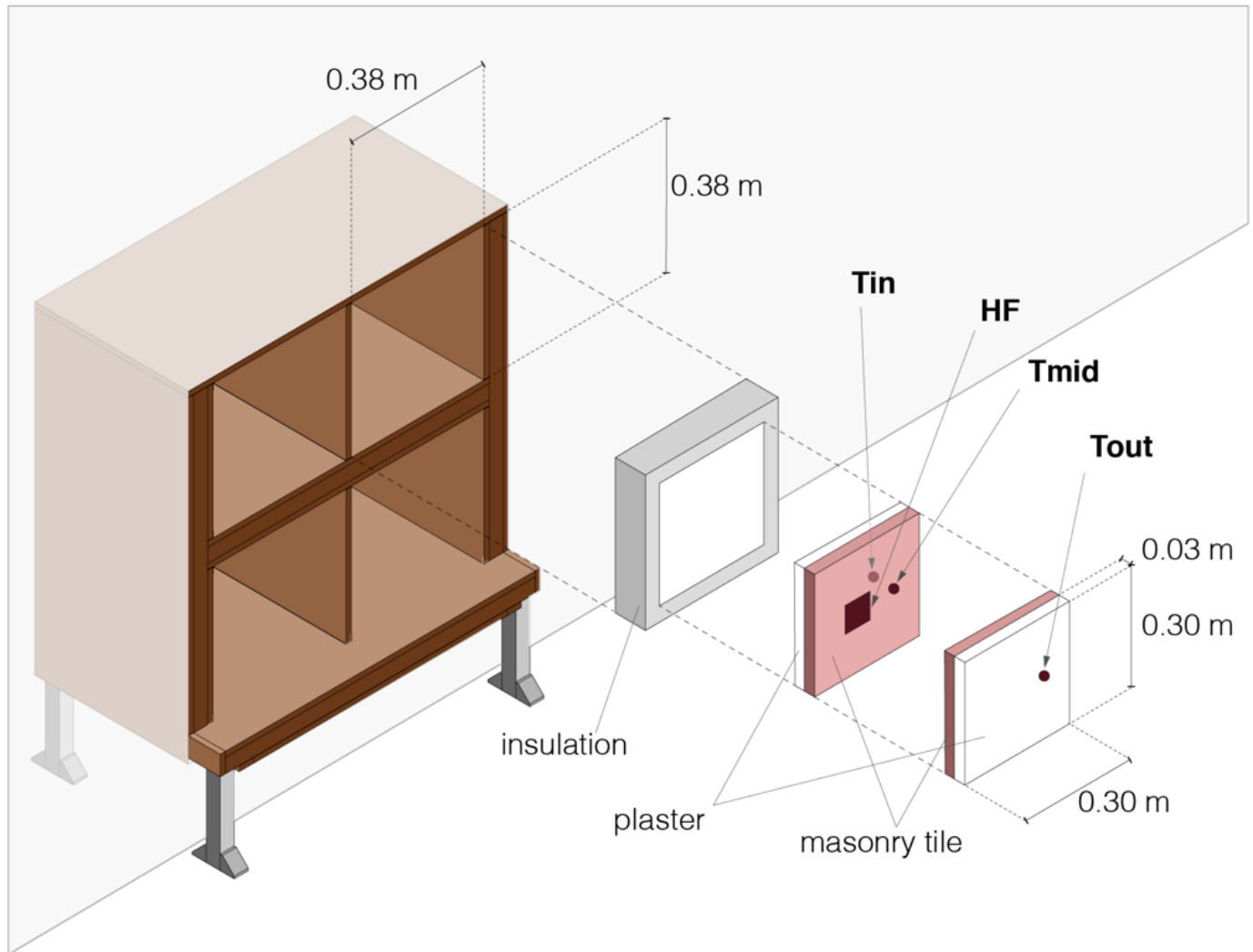


Figure 4

Schematic representation of the set up: materials and sensors' position

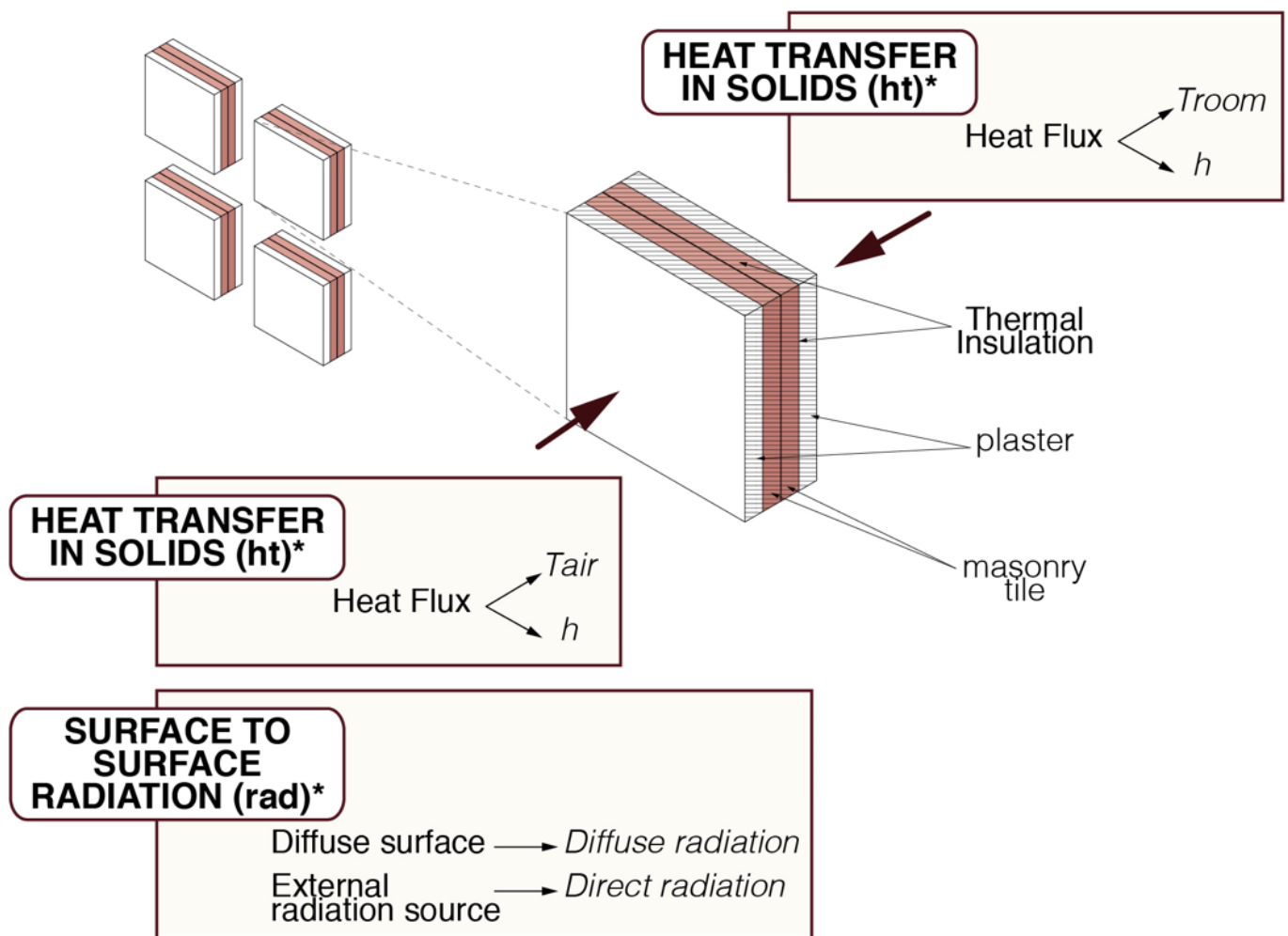
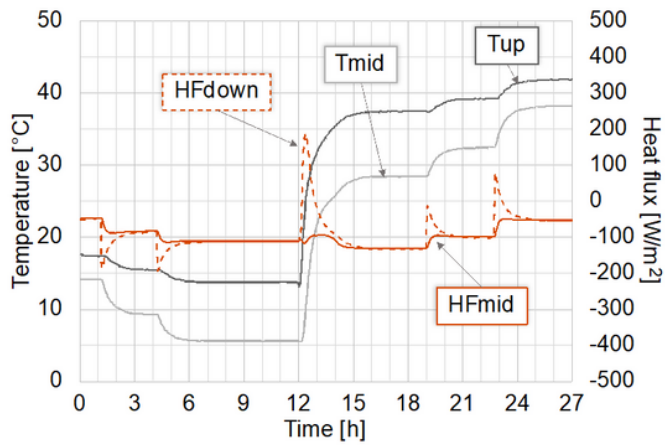
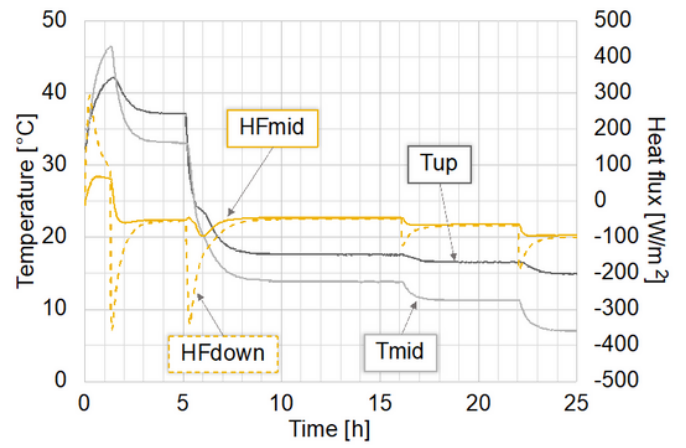


Figure 5

Geometry of the plaster samples implemented in COMSOL Multiphysics and conditions applied on the surfaces



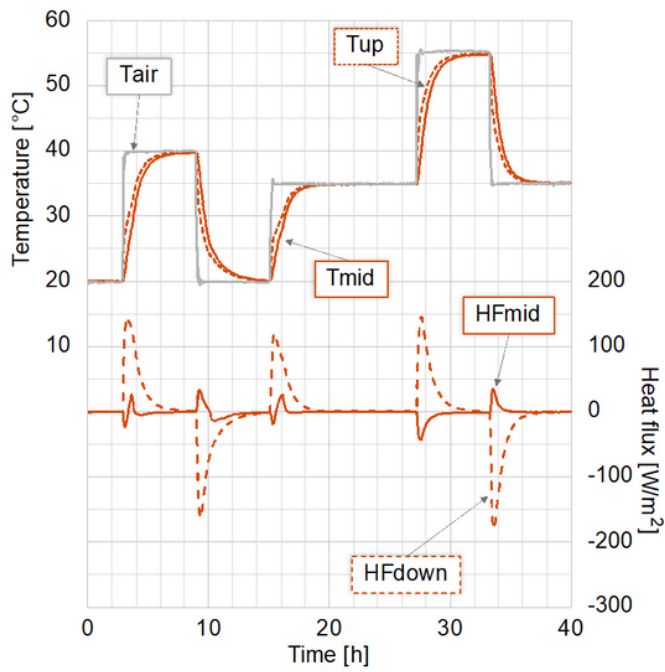
a



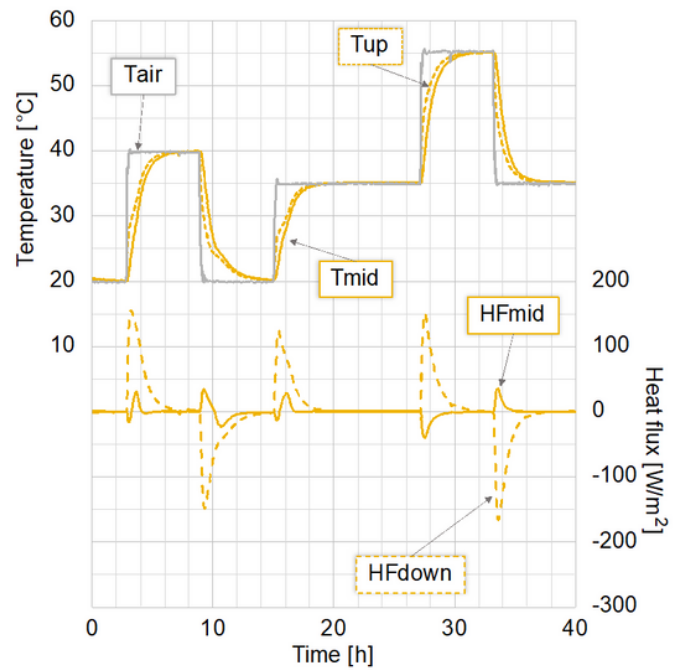
b

Figure 6

Data collected during the steady-state test for (a) AS28 and (b) TK27



a



b

Figure 7

Data acquired during the two-step ramp for (a) AS28 and (b) TK27

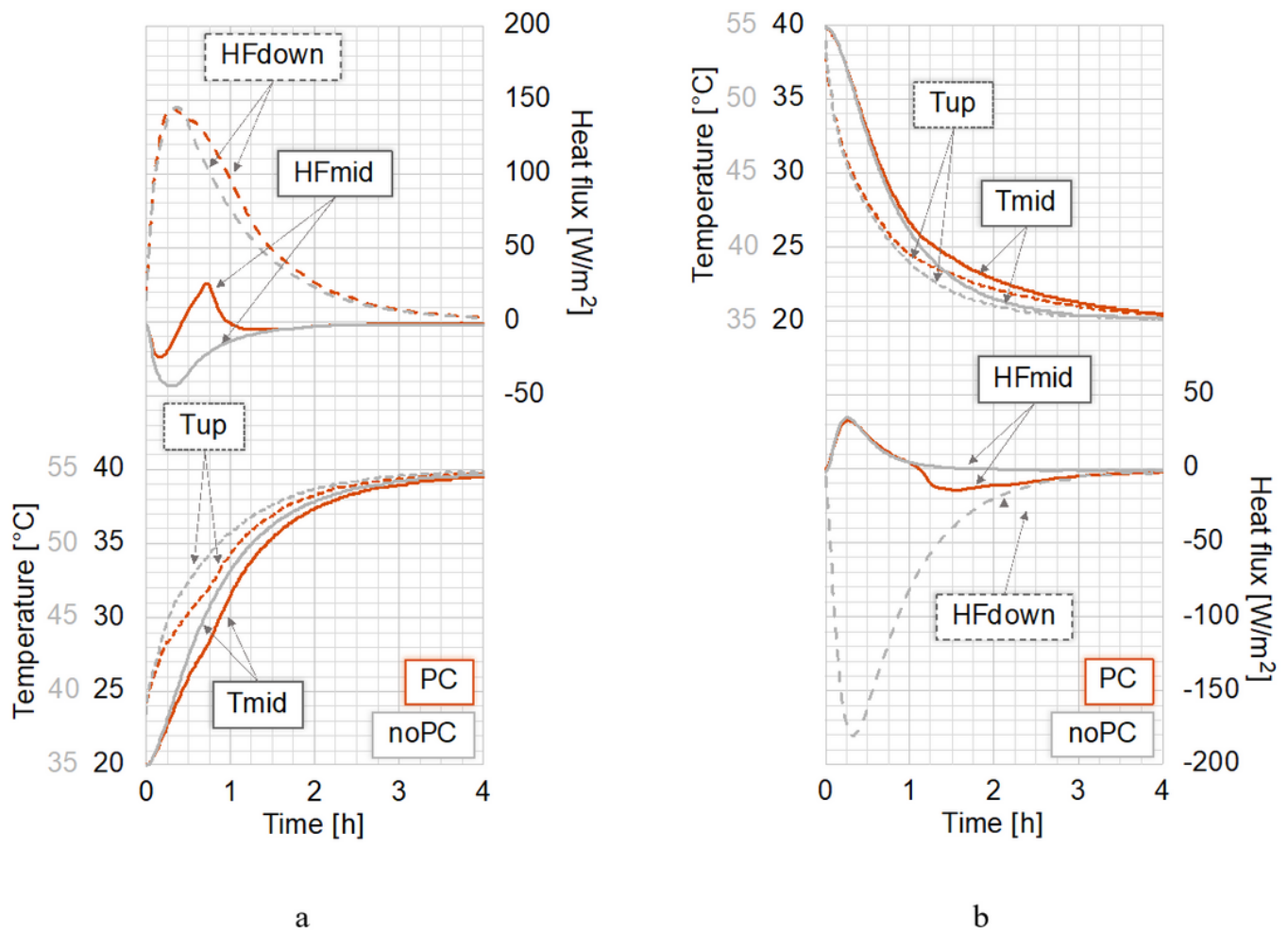


Figure 8

Detailed data collected during (a) heating and (b) cooling of AS28 comparing phase change ramp to the no-phase change one. Grey left-axis numbers are referred to the no-phase change ramp (noPC)

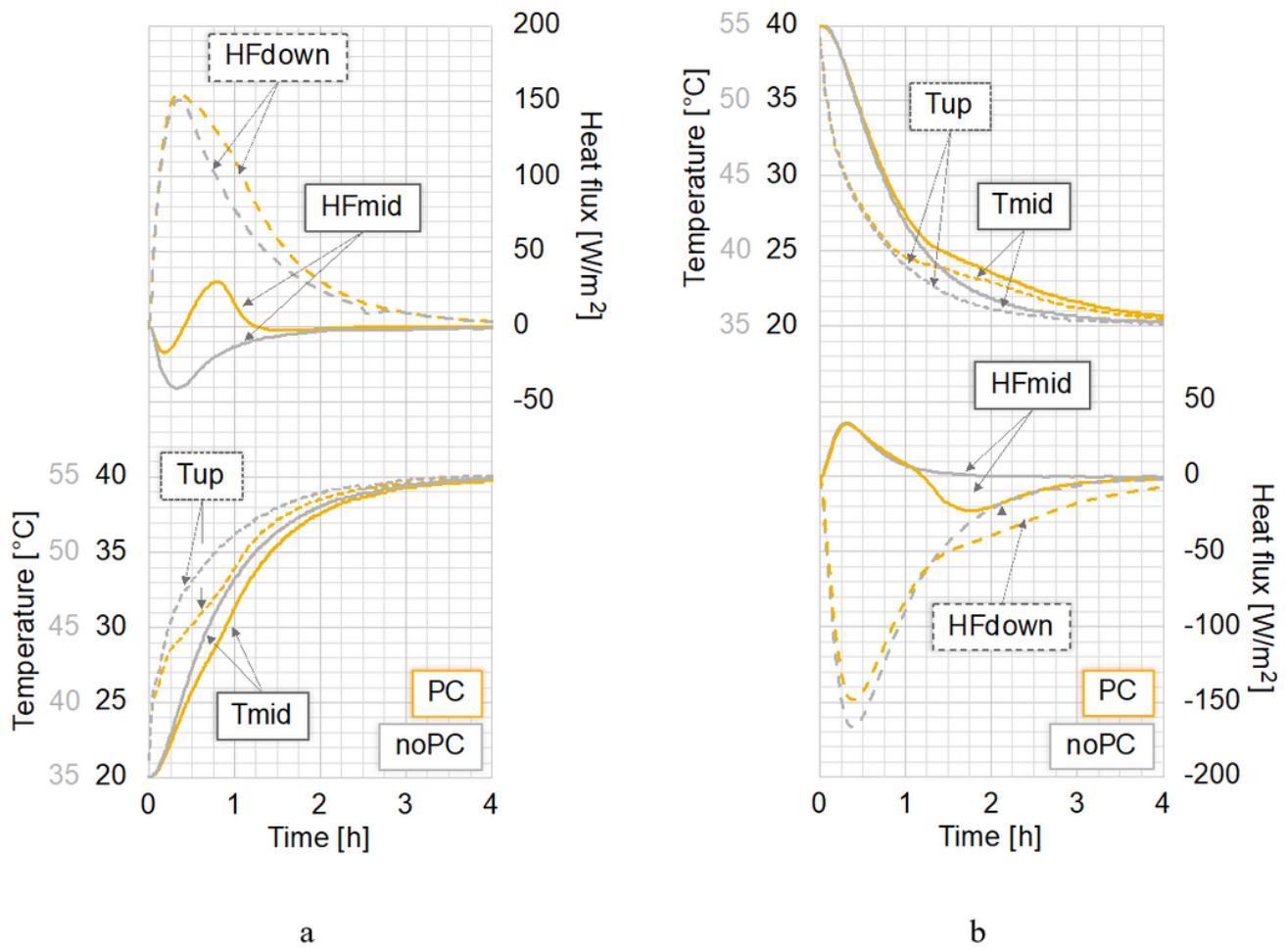
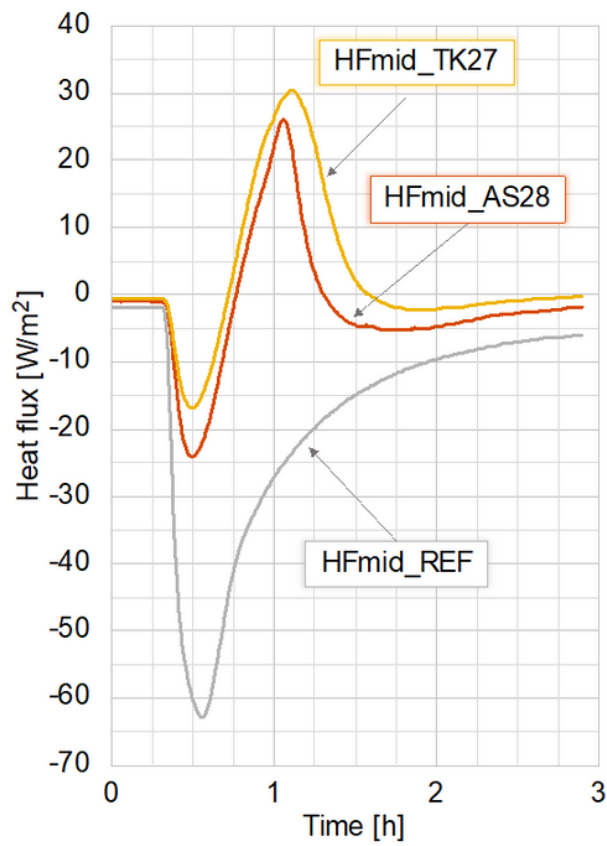
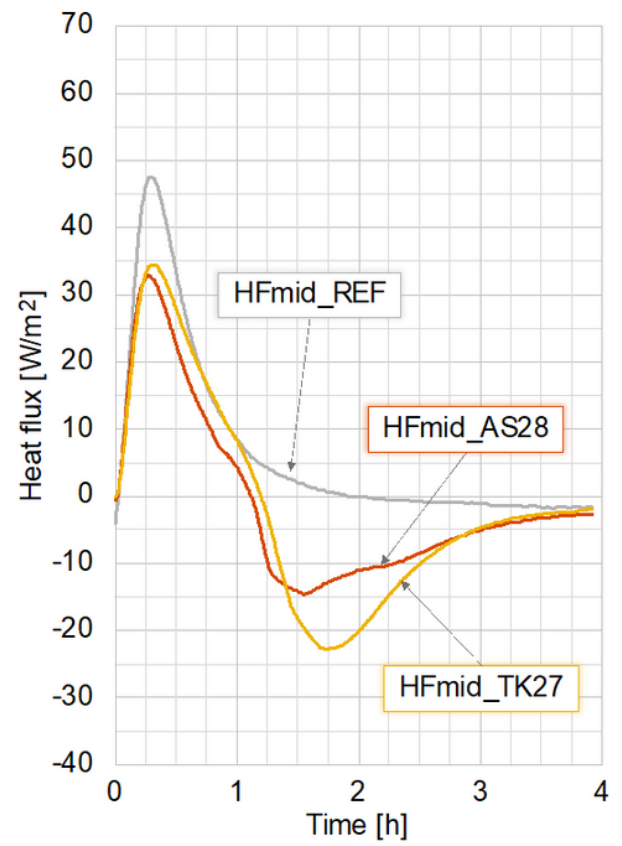


Figure 9

Detailed data collected during (a) heating and (b) cooling of TK27 comparing phase change ramp to the no-phase change one. Grey left-axis numbers are referred to the no-phase change ramp (noPC)



a



b

Figure 10

Comparison of heat fluxes collected during (a) heating and (b) cooling

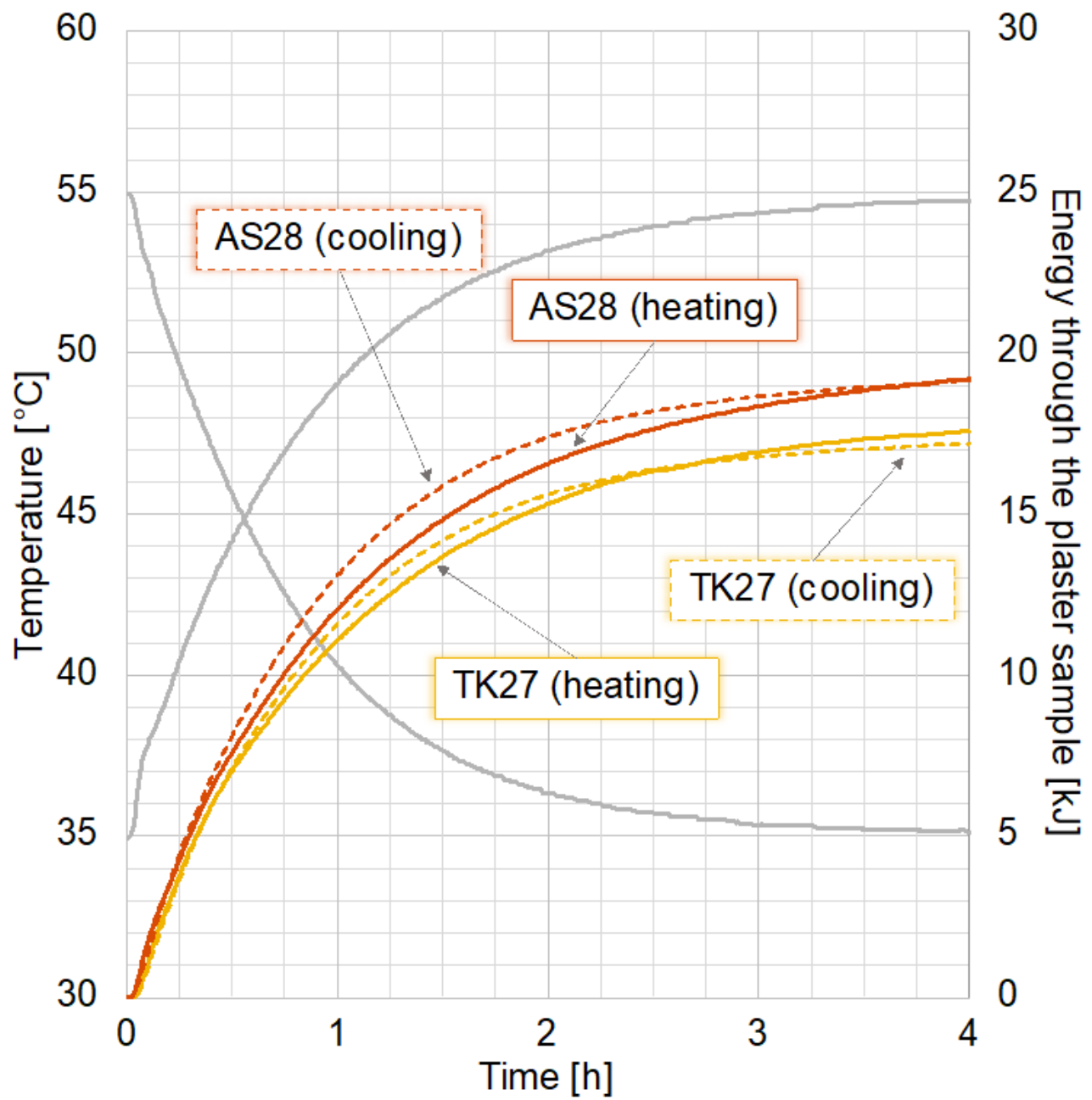
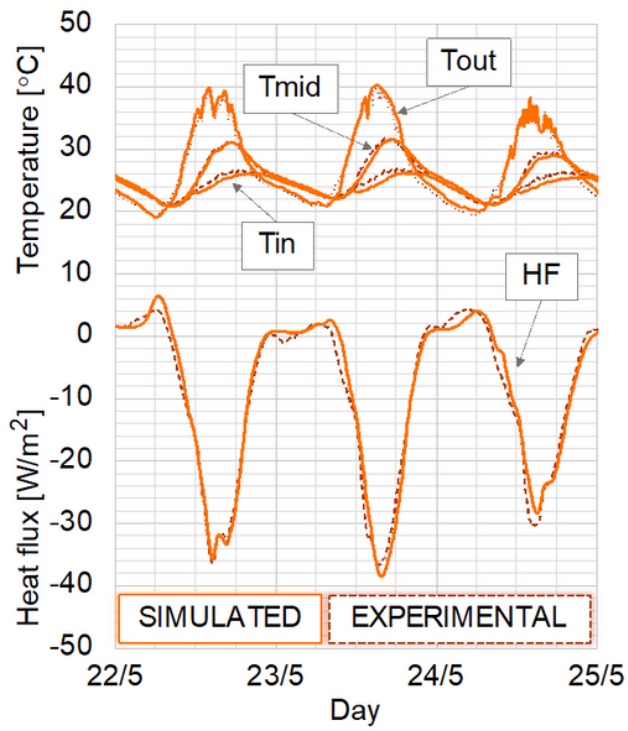
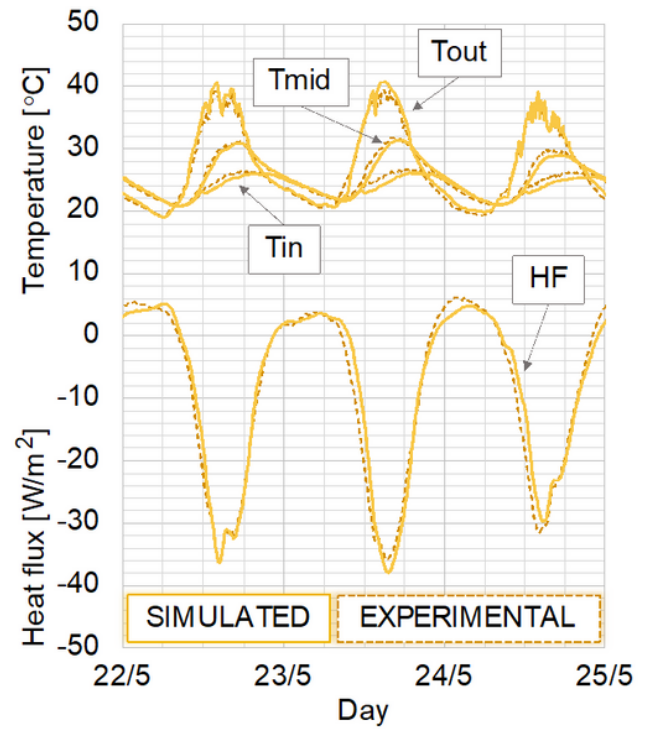


Figure 11

Energy through the plaster samples during heating and cooling



a



b

Figure 12

Comparison between experimental and simulated data for (a) AS28 and (b) TK27